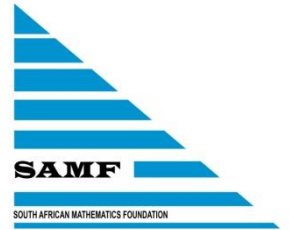




science
& technology

Department:
Science and Technology
REPUBLIC OF SOUTH AFRICA



SIYANQOBA REGIONAL OLYMPIAD TRAINING PROGRAMME 2019

BEGINNERS ASSIGNMENT 2

(Final date of submission: 22_March_2019)

All learners attending Siyanqoba classes are encouraged to submit assignments. All those who submit their solutions will be sent detailed solutions. It will give you invaluable practice for Mathematics Olympiads. This is a great plus. The three assignment marks and your performance in Rounds 1 and 2 will determine whether you will be selected to participate in an all expenses paid Mathematics Camp in Stellenbosch in December.

This is the second assignment of three. PLEASE WRITE YOUR PERSONAL DETAILS AND ANSWERS CLEARLY IN THE ANSWER SHEET on the last page of the assignment. Scan and email ONLY the answer sheet to

siyanqoba.beginner.assignments@gmail.com

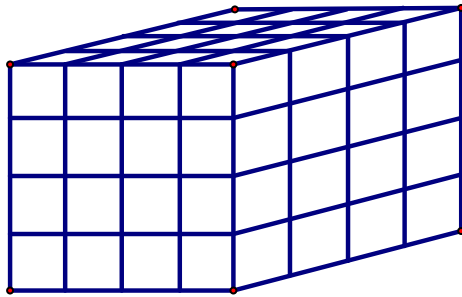
Alternatively, you may hand your answer sheet to your coach who will scan and send it to me.

AS IN ALL OLYMPIADS, CALCULATORS ARE NOT TO BE USED.

1. [4 marks] In a list of six consecutive positive integers the sum of the three smallest numbers is N and the sum of the three largest numbers is M . What is the value of $M - N$?
2. [4 marks] Which number from the set $\{-2; -1; 0; 1; 2\}$ is the smallest value of n for which $3^n \cdot 3^3$ is a perfect square?
3. [4 marks] If $x + y = 2$ & $xy = \frac{1}{2}$, determine $\frac{1}{x} + \frac{1}{y}$.
4. [4 marks] If $f(x) = \left(\frac{1}{2}\right)^x$, what is the value of $\frac{f(x+1) + f(x)}{f(x)}$?
5. [4 marks] A palindromic number is one that reads the same when its digits are reversed. For example, 16233261 is a palindromic number. What is the largest palindromic 8-digit number that is exactly divisible by 36?

6. **[5 marks]** An equilateral triangle and a regular hexagon have equal perimeters. What is the ratio of the area of the triangle to the area of the hexagon?
7. **[5 marks]** Given that a, b, c and d are positive integers with $a < 2b, b < 3c, c < 4d$ and $d < 2$, what is the largest possible value of a ?

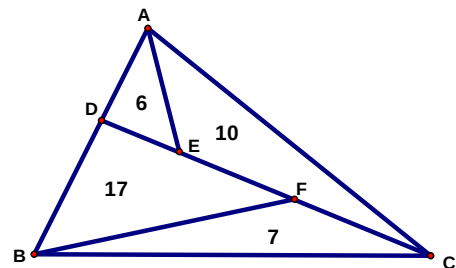
8. **[7 marks]**



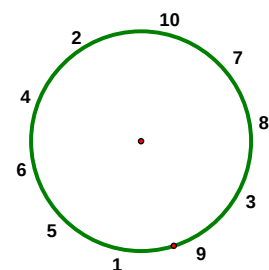
A wooden cube with side n units, $n > 2$, is painted blue all over. It is then cut into n^3 identical small cubes. If the number of small cubes with just one face painted blue is equal to the number of small cubes with no faces painted, find the value of n .

(We have drawn the case where $n = 4$ on the left)

9. **[7 marks]** D is a point on side AB of triangle ABC . E is a point on CD and F is a point on CE . The areas of triangles AED, AEC, BFD and BFC are 6, 10, 17 and 7 respectively. What is the area of triangle BEF ?



10. **[8 marks]** Ten people are seated around a circular table. Each one thinks of a number and then whispers it into the ear of the neighbour on either side of them. Then every person says out loud half of the sum of the two numbers they have heard. The numbers announced are indicated in the diagram on the right. What was the number thought of by the person who announced 6?



11. **[8 marks]** What is the minimum number of integers that must be removed from the first 50 positive odd integers so that the sum of remaining integers is 2019? (**Hint:** the sum of the first n positive odd numbers is n^2)

ANSWER SHEET: ASSIGNMENT 1 (DUE DATE: 25 MARCH, 2019)

Siyanoqoba Centre (Circle one of the following:

UP TUT UKZN BLM EMS RHODES UL UNIVEN/TENGWE WITS ALGOA
SPU NW

Surname:

Name:

Gender:

Race:

Grade:

School:

Cell no:

E-mail address:

QUESTION	YOUR ANSWER	MAXIMUM MARK
1		4
2		4
3		4
4		4
5		4
6		5
7		5
8		7
9		7
10		8
11		8
TOTAL		60

Send to: siyanqoba.beginner.assignments@gmail.com